



LAB -4 DHANUSH.V 1RUA24SCS0031

Lab 4 Handle imbalanced datasets using SMOTE

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In [4]: import pandas as pd
import matplotlib.pyplot as plt
from sklearn.datasets import make_classification
from collections import Counter

X, y = make_classification(
    n_samples=1000,
    n_features=2,
    n_redundant=0,
    n_clusters_per_class=1,
    weights=[0.90],
    random_state=42
)

print(f"Original class distribution: {Counter(y)}")

from imblearn.over_sampling import SMOTE

smote = SMOTE(sampling_strategy='minority', random_state=42, k_neighbors=5)

X_resampled, y_resampled = smote.fit_resample(X, y)
print(f"Resampled class distribution: {Counter(y_resampled)}")

fig, (ax1, ax2) = plt.subplots(1, 2, figsize=(12, 5))

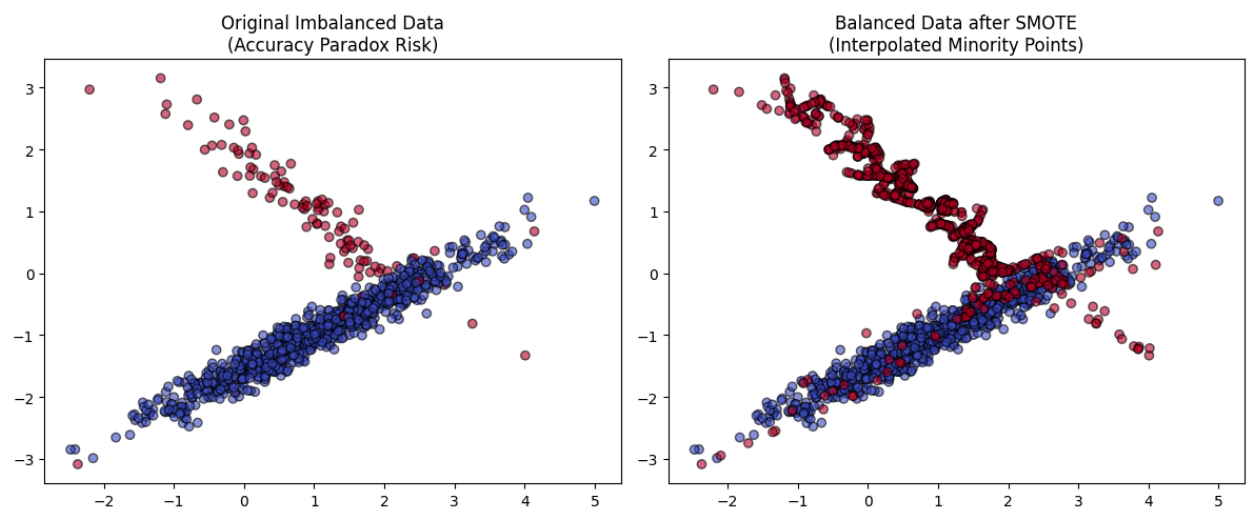
ax1.scatter(X[:, 0], X[:, 1], c=y, alpha=0.6, cmap='coolwarm', edgecolors='k')
ax1.set_title("Original Imbalanced Data\n(Accuracy Paradox Risk)")

ax2.scatter(X_resampled[:, 0], X_resampled[:, 1], c=y_resampled, alpha=0.6, cm
ax2.set_title("Balanced Data after SMOTE\n(Interpolated Minority Points)")

plt.tight_layout()
plt.show()
```

Original class distribution: Counter({np.int64(0): 896, np.int64(1): 104})

Resampled class distribution: Counter({np.int64(0): 896, np.int64(1): 896})



In []: